SWAPNIL MITRA

Software Engineer, Integration & Applied AI

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Software engineer with 3+ years integrating heterogeneous enterprise systems (REST, SOAP, SQL, RPA, insurer APIs) and leading delivery as technical lead from architecture through to production. Builds LLM agent tooling and retrieval in Python, TypeScript, and C#: a multi-protocol tool layer for agents, hybrid-search RAG, and agent memory.

TECHNICAL SKILLS

Languages
Python, TypeScript, C#, SQL, Rust

AI & Data
Claude API, OpenRouter, LLM tool calling, RAG, pgvector, hybrid retrieval, MCP, PostgreSQL, SQL Server, MongoDB

Cloud & Integration
Azure (Functions, API Management, Logic Apps, Entra ID), REST/SOAP, OAuth2/OIDC, MSAL, Microsoft Graph API, RabbitMQ, Docker

Frameworks & Tools
.NET 8, FastAPI, Next.js, React, GitHub Actions, Prometheus, pytest, Git

PROFESSIONAL EXPERIENCE

Adrosonic | **Research and Innovation Engineer** · *Jul 2023 – Oct 2026*

Insurance Broker Management Platform

- After RPA bots re-keying broker answers into insurer websites kept failing, architected an async Azure integration layer to onboard a provider with direct insurer integrations; results return by webhook.
- Cut integration development ~6 weeks to ~2 with a JSON transformation engine on Azure Functions; 99.9% delivery.
- Technical Lead (team of 4) on a multi-tenant broker platform (.NET 8, TypeScript, SQL Server, MongoDB on Azure); designed its Entra ID SSO and cancellation and amendment flows, with their middleware payloads and design.

Policy Administration Integration

- Led a caller-agnostic API giving one contract over several policy administration systems, with async writes.

Release Engineering

- Led a team building a shared, versioned GitHub Actions CI/CD framework for .NET, Functions, and Logic Apps.

Email Filing & Document Management

- Migrated a legacy COM Outlook add-in to a React web add-in with MSAL sign-on and offline caching; 10,000+ users.
- Cut average message latency ~40% moving an email filing and SharePoint pipeline to .NET Core and RabbitMQ.

PROJECTS

- **Flight Recorder** · *Nov 2026* Python, OpenAI + Anthropic SDKs, LangChain/LangGraph, OpenTelemetry, pytest
  - Built and published to PyPI a Python package that records every model and tool call of an AI agent run to JSONL (tokens, cost, latency) and replays runs as pytest regression tests in CI.
  - Added pre-call budget caps and model routing, plus approval gates that hold irreversible tool calls, fail closed, and export an audit trail; recording adds under 2 ms per call.
- **Heptapod** · *Sep – Oct 2026* Python, Anthropic + OpenRouter tool calling, FastAPI, SOAP, OAuth2, Docker Compose
  - Built a Python layer giving LLM agents one tool interface over REST, SOAP, and SQL systems, handling API key, OAuth2 client credentials, and HTTP basic auth; a new system is one adapter class plus one config entry.
  - Made adapters return structured results instead of raising on HTTP errors, SOAP faults, timeouts, and 429s so the model can self-correct; tested on three FastAPI mocks in Docker Compose with pytest in CI.
- **MemVault** · *Aug 2026* Rust (edition 2024), Cargo workspace, usearch, BM25, MCP, gRPC, PyO3
  - Built a local-first Rust memory engine for AI agents: hybrid retrieval, hash-chained ledger, provable deletion.
- **Colloquium** · *Jul – Aug 2026* Tauri (Rust), React, TypeScript, WebRTC, SQLCipher, Zustand
  - Co-built a Discord-style P2P desktop app (macOS, Windows) with no backend: chat, voice/video rooms, screen share and watch parties run over WebRTC; a public broker only introduces peers.
  - Built the chat and call UX in React/TypeScript: Discord-style composer with markdown and emoji, drag-and-drop and pasted attachments with progress, desktop notifications, device selection, and a floating call window.

EDUCATION

Indian Institute of Technology, Kharagpur | **B.Tech, Biotechnology and Biochemical Engineering** · *May 2023*

EXTRA-CURRICULARS

- Associate Manager, Entrepreneurship Cell, IIT KGP: organised the Global Entrepreneurship Summit 2021.